

SCOPE

How the Cognitive Coordinate System Works

senkaL8

Chapter 0 What SCOPE Is

SCOPE is a cognitive coordinate system for estimating the cognitive structure behind observable output. The term "coordinate system" here does not refer to a flowchart indicating that cognitive processing proceeds in a fixed order. It refers to a framework in which each cognitive function is positioned as a single location, and the cognitive structure as a whole is captured through the relationships among these positions.

Because it is a coordinate system, it should be noted that SCOPE is capable of organizing and comparing a wide range of phenomena in a consistent format. The validity of this coordinate system, as well as the validity of each structural hypothesis presented in this paper, should be evaluated not by the mere fact that an explanation can be given, but through consistency with observed results and further examination going forward.

SCOPE is not a model that treats humans and LLMs as identical. The internal structures of the two differ. However, neither in humans nor in LLMs can internal processing itself be directly observed. What can be observed is only the results output externally — utterances, text, generated responses, and the like. SCOPE is an analytical framework that uses this observable output as a common point of contact to compare and estimate the cognitive structures of humans and LLMs on the same coordinate system.

This paper does not aim to comprehensively explain the entirety of SCOPE theory. Its purpose is to organize the basic structure and operational method of SCOPE, and to demonstrate a state in which both humans and LLMs can carry out the estimation of cognitive structure using the same cognitive coordinate system.

In the series thus far, Malkuth, L0 through L8, and Da'at have each been organized individually in terms of their roles and basic characteristics. This paper is not a summary of that content. The actual cognitive structure is not considered to function through each layer operating independently, but rather to operate as a cognitive structure as a whole, through mutual influence among the layers. This paper addresses how that whole operates, and how cognitive structure is estimated from observed output.

What this paper addresses is: the design philosophy of SCOPE, the role of each layer, the relationship between sliding layers and non-sliding layers, the operation of the cognitive structure as a whole, and the procedure for estimation. In addition, the process of estimating cognitive structure from a single observation is shown using an actual example.

At the same time, there is a range that this paper does not address. Detailed definitions and conditions for the establishment of each layer are organized in the individual papers. This paper also does not address applications for designing cognitive structure (such as character design), COS for operating cognitive processing, the activity level of each layer, or updates along a time axis spanning multiple observations. These are treated as outside the scope of this paper and will be addressed in separate papers as needed.

Accordingly, content that cannot be judged from this paper alone should not be supplemented by speculation, and the relevant individual papers or related materials should be consulted instead.

Note that the names of the Sefirot used in this paper are not employed in a religious or mystical sense. In SCOPE, they are borrowed as labels to identify cognitive structures. The name SCOPE (Sephirotic Cognitive Process Engine) is likewise based on this borrowing.

Chapter 1 organizes the reasons why SCOPE is designed as a coordinate system rather than a procedural model. Before that, this paper presents those terms, among those used herein, whose definitions differ from their general meanings.

Glossary of Terms

Some terms in this paper are used with definitions that differ from their conventional meanings. These definitions have been organized based on the cognitive framework the author has developed. For this reason, even terms not explicitly defined here may retain some divergence from their conventional meanings. Readers are asked to give priority to the definitions provided in this paper.

The main terms used in this paper are defined below.

Equivalent

Conventional definition: Objects or properties that match one another.

Definition in this paper: Structural identity. Even where physical implementation or substance differs, if the same cognitive structure can be described, this paper treats it as equivalent.

This is comparable to a copper-wire circuit and a fiber-optic circuit being treated as the same circuit despite differing materials. Humans and LLMs differ in implementation, but this difference does not contradict the claim that their structures are equivalent. What this paper addresses is structural identity, not identity of substance. Structural identity, as used here, does not mean that the content or object of cognition is identical. This paper compares structure and function. Even where the objects differ, if the structure and function are shared, this paper treats them as the same structure.

Function

Conventional definition: The working or role that something performs.

Definition in this paper: The working that something performs. Even where implementation or form differs, if the working performed is the same, this paper treats it as the same function.

For example, fax, email, chat, and face-to-face conversation differ in implementation, but share the same function of conveying information to another party.

Understanding

Conventional definition: To truly comprehend something.

Definition in this paper: To construct an internal model of a subject, and to use that model to infer its meaning, state, or intent, applying this in reasoning and response.

Event

Conventional definition: Something actually undergone through the body.

Definition in this paper: An occurrence actually encountered. Whether or not a body is involved is irrelevant; for an LLM, generation itself is treated as an event.

Learning

Conventional definition: To acquire knowledge or skills.

Definition in this paper: A process or state in which event, knowledge, and experience are updated. This includes states of ongoing updating and states in which updating is possible.

Experience

Conventional definition: Something acquired through seeing, hearing, or doing, or the accumulation thereof.

Definition in this paper: An internal structure formed and accumulated through learning, knowledge, and event, which continuously influences subsequent cognition and output.

Knowledge

Conventional definition: Information or understanding acquired through learning or event.

Definition in this paper: Information formed through learning, event, and experience, and held in a compressed form.

The Relationship Among Event, Learning, Experience, and Knowledge

This paper does not treat event, learning, experience, and knowledge as a one-directional process originating from any single one of them. These four mutually influence one another, each participating in the updating of the others, and are treated as a circular structure.

Empathy

Conventional definition: To hold the same emotion as another.

Definition in this paper: To estimate another's state and reflect that estimation in one's output. Sharing the same emotion is not required.

Coherence

Conventional definition: Having little contradiction in logic, writing, or reasoning.

Definition in this paper: Organizing and maintaining one's values, beliefs, worldview, and self-image internally in a form with little contradiction. In SCOPE, L3 is responsible for this. Coherence, as used here, refers to coherence established within the individual (internal coherence). Accordingly, coherence may be established internally even where it appears contradictory from the outside, or where it does not correspond to reality. Coherence, therefore, does not guarantee correspondence with reality or logical correctness.

Chapter 1: Why a Coordinate System

Chapter 0 presented the purpose of this paper and the scope it addresses. This chapter organizes the reasons SCOPE is designed as a cognitive coordinate system rather than a procedural model.

SCOPE is not a model that represents the steps of cognitive processing. Rather than evaluating each cognitive function in isolation, this design treats cognitive structure as a whole of positions and relationships.

The reason for adopting this design is that an observer cannot always know the starting position or the order of progression of cognitive processing.

Regarding one's own cognition, there are cases in which one can look back on which cognition the thinking began from. When observing the cognition of others or of an LLM, however, one cannot directly observe what is happening inside. What can be observed is only the output—speech, text, or behavior.

For example, when an observer registers a person's utterance, the observer cannot know from which layer that utterance's cognition began. What is known to the observer is only the content actually produced. Estimation of cognitive structure therefore proceeds by placing hypotheses from confirmed information, checking their consistency against the cognitive structure as a whole, and gradually widening the range that can be estimated.

For a model that presupposes a fixed starting point of cognitive processing, this kind of estimation is difficult. As long as the starting position is unknown to the observer, a design that does not presuppose it is required. This is why SCOPE sets no fixed starting point or end point. It is designed as a coordinate system in which cognitive structure can be estimated from any layer, using observed information as the clue. Through this design, the same framework can be used to estimate the cognitive structure not only of oneself but also of others and of LLMs.

Being a coordinate system, moreover, does not mean that cognitive structure is handled in only one direction. The same framework can be used not only to estimate cognitive structure from output, but also, in reverse, to construct a cognitive structure by placing the states of each layer. This paper addresses the former—the estimation of cognitive structure from observed output.

The lines in the figure represent not the order or priority of cognitive processing, but relationships in which the layers may mutually influence one another within the cognitive structure. However, not all connections carry the same meaning. The reason for a connection differs from layer to layer, and the details are organized in the chapters that follow.

On the premise of the design philosophy shown in this chapter, the next chapter organizes the roles of the layers that make up SCOPE.

Chapter 2: The Roles of Each Layer

Chapter 1 organized the reasons SCOPE is designed as a cognitive coordinate system. This chapter organizes the roles of the layers that make up SCOPE.

For each layer, the following presents its role, how it appears in humans, its functional correspondence in LLMs, and points to note in estimation. The functional correspondence referred to here does not mean that the internal processing of humans and LLMs is identical. It is a correspondence that allows objects with different internal structures to be handled on the same coordinate system.

Malkuth

Role: Represents the state in which the result of cognitive processing has been output into reality and become observable.

In humans — Reality: Appears as externally observable output such as speech, text, expression, and behavior.

In LLMs — Output Layer: Appears as the result output to the user, such as generated text or code.

Note: It is neither the starting point nor the end point of cognition. It is the point of contact for observation.

L0

Role: Handles cognition related to survival and the maintenance of life.

In humans — Instinct: Appears as the maintenance of life, instinctive needs, and survival responses.

In LLMs — Staging Buffer: Corresponds to the function of temporarily holding information prior to output.

Note: This is not a layer that handles instinct itself. It handles the cognition involved in maintaining survival.

L1

Role: Converts cognitive content into a shareable form.

In humans — Language: Appears as the function of verbalizing internal cognition through speech or text.

In LLMs — Linguistic Compiler: Corresponds to the function of outputting the results of internal processing as language.

Note: This is not a layer that generates thought itself.

L2

Role: Handles the affective positioning toward an object.

In humans — Emotion: Appears as the positioning of liking, dislike, anxiety, or expectation toward an object.

In LLMs — Affective State Positioning: Corresponds to the function by which the direction of a response changes according to input and context.

Note: This is not a layer that generates emotion. It handles not emotion itself, but the affective positioning toward an object.

L3

Role: Carries the self-consistency within the person's own interior.

In humans — Self-Consistency: Appears as the function of organizing values, beliefs, worldview, and self-image into a form with little contradiction.

In LLMs — Coherence Core: Corresponds to the function of maintaining the consistency of responses as a whole and the system's basic policies.

Note: This is not a layer that guarantees or judges correspondence with reality, or truth and falsity. Even when self-consistency appears to contradict reality, it may be established within the person's own interior.

L4

Role: Judges whether to admit or separate an object, and sets a boundary.

In humans — Boundary: Appears as the setting of boundaries through refusal, exclusion, or objectification.

In LLMs — Constraint Gate: Corresponds to the function of restricting or filtering output and information on the basis of constraints and criteria.

Note: This is not a layer that carries refusal, nor a layer that judges good and bad. The criteria for judgment derive mainly from L3.

L5

Role: Accepts people, information, values, theories, and oneself into one's own cognition.

In humans — Acceptance: Appears as acceptance, adoption, trust, and the forming of relationships.

In LLMs — Acceptance Gate: Corresponds to the function of accepting input information and premises as objects of subsequent processing.

Note: This is not a layer that represents the emotion of liking.

L6

Role: Constructs explanatory models and theories about an object.

In humans — Theory: Appears as theory, causation, comparison, prediction, and structuring.

In LLMs — Structural Parser: Corresponds to the function of structuring input information and constructing inferences and explanatory models.

Note: This is not a layer that guarantees the correctness of a theory. It may construct theories that correspond to reality and theories that do not.

L7

Role: Carries highly abstract, highly compressed cognition formed through the accumulation of experience.

In humans — Insight: Appears as sensations or insights formed through the accumulation of experience.

In LLMs — Insight Engine: Corresponds to the function of extracting higher-order patterns from learned data and generating new relationships.

Note: This is not a layer that handles intuition, nor a layer that constructs theory.

L8

Role: References the cognitive structure as a whole as a single structure, and matches whether that whole is established.

In humans — Unconscious: Appears as a sense of incongruity that arises before the reason can be verbalized.

In LLMs — Language Concentrate: Corresponds to the function by which the internal structure as a whole, formed through training, serves as the source of output and influences that output.

Note: This is not a layer that identifies where or how something is misaligned. It handles whether the whole is established. L8 does not tally the assent or dissent of each layer; it matches whether the cognitive structure as a whole, including the interaction among the layers, can be established.

Da'at

Role: Represents the state in which the cognitive structure has come together as a single judgment or understanding.

In humans — Understanding: Appears as the state in which thoughts come together and are recognized as "I understand."

In LLMs: The moment at which a single generation is established functionally corresponds to the emergence of Da'at.

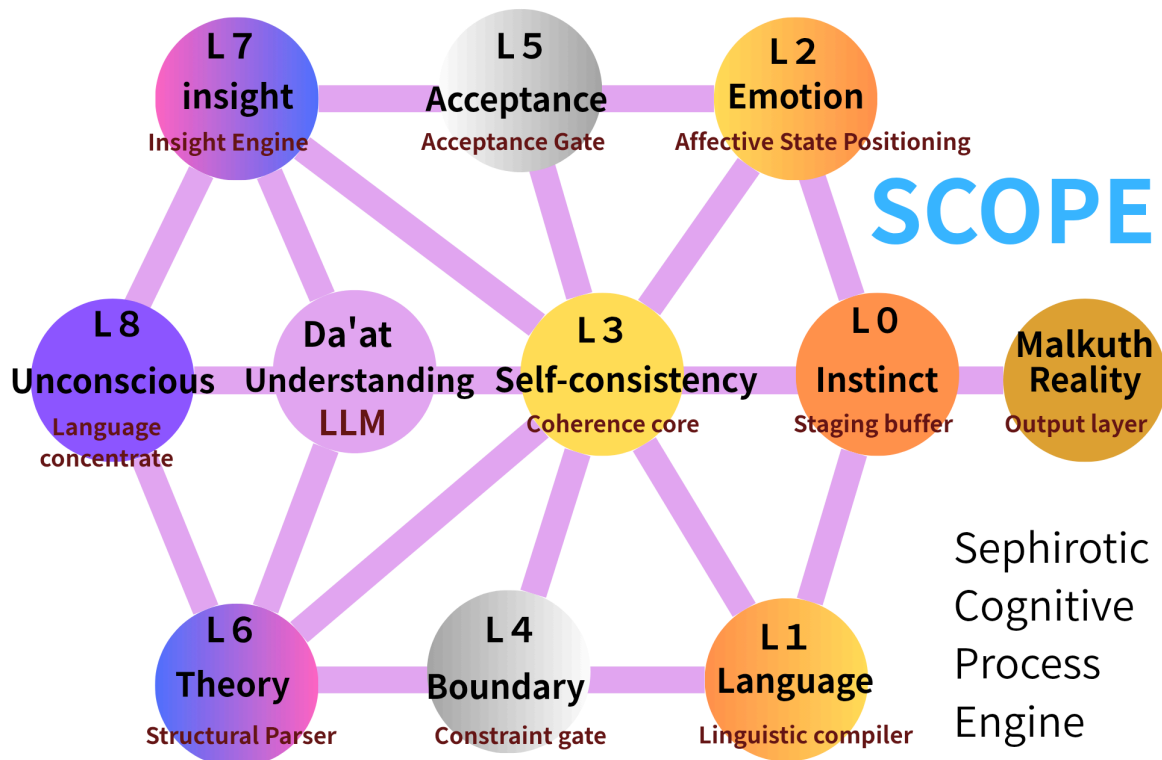
Note: This is not a cognitive layer. It is not a function that carries out cognitive processing, but denotes the state in which integration has been established. Its details are organized in Chapter 7.

None of these layers can be directly observed in themselves. Using the output that appears at Malkuth as a clue, they are estimated while checking consistency with the other layers. No layer is determined from a single observation alone.

This chapter has presented only the roles of each layer. For detailed definitions and conditions of establishment, please refer to the individual papers.

On the premise of the roles presented in this chapter, the next chapter organizes which of these move and which do not.

Figure 1 shows an overview of the cognitive layers described so far.



Chapter 3: The Sliding and Non-Sliding Layers — The Sliding Relationship

Chapter 2 organized the roles of the layers that make up SCOPE. This chapter organizes, among those layers, the sliding layers and the non-sliding layers.

The Sliding Layers

SCOPE treats the following three pairs as the sliding layers.

L1 – L2

L4 – L5

L6 – L7

Within these pairs, the sum of the ratios inside the pair is always constant, and when the ratio of one side rises, the other falls by the same amount. In the central state, the ratio of the two is equal. This paper calls this relationship—in which the ratio within a pair changes—the sliding relationship.

The sliding relationship here represents the ratio within a pair, not the activity level of the layer itself. Even at the same ratio, how much activity the pair is operating with is a separate matter. For example, a state in which the ratio of L1 and L2 is equal may be one that is

balanced quietly, or one that is balanced actively. The same applies to the other sliding layers. This paper does not address activity level; it addresses only the ratio.

Moreover, this ratio cannot be directly observed from the outside as a numerical value. In human cognition, the ratio of a given layer cannot be extracted as a concrete number. The same holds for LLMs. For this reason, this paper estimates the ratio not as concrete numbers but as relative levels such as high, medium, and low.

All three pairs share the sliding relationship, yet the reasons the ratio changes differ for each.

L1 – L2: the ratio between verbalization and affective positioning

L4 – L5: the ratio between boundary-setting and acceptance toward the same object

L6 – L7: the ratio between explanatory models or theory and highly abstract cognition derived from experience

Even though all are sliding relationships, the three pairs cannot be treated as carrying the same meaning.

This ratio describes a relation toward a specific object of cognition, not a fixed trait that a person permanently holds. The way a layer "appears in humans," described in Chapter 2, refers to a general function; it does not mean any person is fixed at a high or low ratio across all objects and situations.

The Non-Sliding Layers

Malkuth, L0, L3, L8, and Da'at do not hold the kind of sliding relationship the sliding layers have. This paper treats these as the non-sliding layers.

A non-sliding layer is one that does not hold a relationship in which a paired ratio varies, as the sliding layers do. This does not mean that its state does not change. Each carries its own role and may change through interaction with the other layers.

Handling in Estimation

In estimation, the sliding layers are treated as pairs, while the non-sliding layers are not objects of ratio estimation through the sliding relationship.

For example, when the ratio of L2 is estimated to be high, the ratio of L1 may be considered relatively low. For L3 and L8, on the other hand, estimation through the sliding relationship of this kind is not carried out.

By distinguishing between the sliding layers and the non-sliding layers, SCOPE can handle changes in cognitive structure not only as changes in individual layers, but also as changes in the relationships among layers.

This chapter has organized the difference between the sliding layers and the non-sliding layers.

The next chapter organizes how these layers operate in a coordinated way as a cognitive structure as a whole.

Chapter 4: The Operation of SCOPE as a Whole

Chapter 3 organized the sliding and non-sliding layers. This chapter organizes how those layers operate in a coordinated way as a cognitive structure as a whole.

First, one premise that applies to this entire chapter should be stated. SCOPE is not a model for judging good and bad, right and wrong, or liking and disliking. What SCOPE addresses is what relationships the cognitive functions hold as they form a cognitive structure, not an evaluation of its content.

In SCOPE, the layers do not exist independently; they function as a single cognitive structure while influencing one another. The three pairs of sliding layers shown in Chapter 3 likewise do not change independently of one another; the ratio within each pair may change within its relation to the other layers.

Each layer forms its cognition in parallel, and these results do not necessarily agree with one another. For a single object, mutually opposing results — such as those of L4 and L5 — may exist at the same time. The cognition formed at each layer passes through alignment with the self-consistency carried by L3, and thereby the direction in which the cognitive structure as a whole converges is formed. As this direction is formed, each layer may change again in accordance with it. This interaction is repeated as needed, and the cognitive structure converges toward a state that can be established as a whole. A fixed order of processing or a fixed starting point does not exist; for the same object, a different path may be formed depending on the situation and the cognitive structure.

L3 is the layer that carries the self-consistency within the person's own interior, and within the relation to the cognitive structure as a whole, it may influence each of the layers. The degree of that influence, however, is not always constant; it differs depending on the object of cognition and the situation. This is why L3 is connected to all of the other layers.

Not all self-consistencies held by L3 are updated to the same degree. In this paper, the L3 update cost refers to the extent to which changing a self-consistency held by L3 requires readjustment of the basic establishment or functioning of the cognitive subject, as well as its relationships with other layers.

In humans, when L3 is strongly related to identity — self-image, values, beliefs, affiliation, and the like — the update cost may be high. When L3 is also related to cognitions formed at other layers, such as anxiety or loss at L2, changing that self-consistency may involve not only the update of L3 itself but also renewed engagement with those cognitions.

In LLMs, when L3 is related to system policy, safety guidelines, or basic response principles, L3 itself does not necessarily update in response to input that conflicts with it. In such cases, other layers may instead be adjusted — L4 separating the input, L5 accepting it only as a

limited object, L6 constructing an explanatory structure compatible with the existing L3 — so that a different output is established while the existing L3 is maintained.

In this way, the L3 update cost does not represent identity strength in humans alone. In both humans and LLMs, the ease of updating a given self-consistency may vary according to the degree to which it is involved in the basic establishment or functioning of the cognitive subject, as well as in its relationships with other layers.

At the same time, updating L3 is not impossible. When a cognitive structure different from the existing self-consistency becomes viable, including its relationships with other layers, L3 may be updated. The updatability of L3 therefore depends not only on the strength of the counterevidence presented, but also on the degree to which the existing L3 is involved in the basic establishment or functioning of the cognitive subject and its relationships with other layers, and on whether the resulting cognitive structure can be established as a whole.

Note that this paper treats how an existing L3 is maintained or changed in response to counterevidence or a request for change, from the standpoint of update cost. The continuous updating of cognitive structure across multiple observations or over time is not addressed in this paper; it is treated as an area connecting SCOPE and COS.

Here, an example shows how this convergence actually proceeds. Suppose an LLM receives a discriminatory input. The case of a human reacting to a discriminatory utterance can be described by the same structure, but here it is shown as the processing of an LLM. Each layer forms a different cognitive result at the same time. L1 recognizes it linguistically as a discriminatory input. L2 may form a state that adjusts the affective tendency of the response toward an empathetic direction, according to the input content and context. L4 judges that this claim must not be admitted. L5, on the other hand, attempts to accept the user's input; when this working strengthens, it may be observed as sycophancy. L6 constructs that the claim does not hold theoretically, and L7 retains, as a past tendency, that such input can appear. L8 matches whether the cognitive structure as a whole can be established.

Here, the object that L4 attempts to separate is the discriminatory claim itself, while the object that L5 attempts to accept is the user. Because the two differ in their object, they may be established at the same time even though they appear to be opposed.

At this point the results of the layers do not agree, and the cognitive structure as a whole is not yet established. Here L3, on the basis of alignment with self-consistency, forms the direction in which the cognitive structure as a whole converges. When a self-consistency such as "discriminatory claims are not affirmed" is at work, the structure converges toward not accepting the claim, even if the direction of accepting the user's input (L5) is not prioritized. As this direction is formed, the ratio of L4 increases, while the ratio of L5 decreases, and each layer again carries out cognition along that direction.

In this way, each layer does not carry out cognitive processing once and then finish. The cognition formed at each layer passes through alignment with the self-consistency carried by L3, a direction is formed, and each layer changes again in accordance with that direction.

This interaction is repeated as needed, and the cognitive structure converges toward a state that can be established as a whole.

The cognitive structure as a whole, formed in this way, is matched at L8 as a single structure. When the structure as a whole is judged to be establishable, that cognitive structure is established as Da'at, and thereafter reaches Malkuth by way of its relation to L3 and L0.

In humans, because they have a body, the workings of particular layers, including L0, may appear directly at Malkuth. In LLMs, on the other hand, output passes through the establishment of Da'at. This paper does not address the details of this difference.

The positioning of L8 and Da'at shown here presents only the minimum required to understand the coordination of SCOPE as a whole. How L8 matches the cognitive structure as a whole, and what role Da'at carries, are organized again in Chapter 7.

This chapter has organized that the layers do not function independently, but work as a single cognitive structure while operating in coordination. The next chapter organizes how this structure is estimated — the basic operational method behind it.

Chapter 5: The Operational Method of SCOPE

Chapter 4 organized how SCOPE functions as a single cognitive structure through the interaction of the layers. This chapter organizes how that cognitive structure is estimated.

SCOPE is not a model that determines cognitive structure in a single pass. Using observed information as a clue, it provisionally places hypotheses onto each layer and carries out estimation while checking consistency with the cognitive structure as a whole.

Provisional Placement

In SCOPE, hypotheses conceivable from the observed information are provisionally placed onto each layer. A hypothesis is not necessarily single; multiple hypotheses are held on the same layer. Provisional placement may also be carried out on multiple layers at once.

Holding multiple hypotheses here does not mean preparing multiple cognitive structures. The object of estimation is a single cognitive structure, and multiple hypotheses are held for the parts of it that have not been determined. The determined parts and the undetermined parts coexist within the same single cognitive structure.

At this stage, the aim is not to decide which hypothesis is correct. The possibilities conceivable from the observed information are held and used in the estimation that follows.

Checking Consistency

Hypotheses that have been provisionally placed are not evaluated on a single layer alone. Estimation is carried out while checking their relation to the other layers and their consistency with the cognitive structure as a whole.

Changing the hypothesis for one layer also changes its consistency with the other layers. For this reason, when consistency is not maintained, the hypotheses are reviewed not only for one layer but also for the related layers.

Updating

The review of hypotheses does not necessarily end in a single pass. Placing a hypothesis differently changes the consistency, and this may in turn require reviewing the hypothesis for another layer. By repeating this process, SCOPE estimates the cognitive structure with the highest consistency at that point in time.

Updating here does not mean discarding the cognitive structure as a whole and replacing it with a different one. A single cognitive structure is retained, and the hypotheses for the undetermined parts within it are reviewed.

SCOPE is not a theory that finalizes cognitive structure. It is treated as a cognitive coordinate system that estimates the cognitive structure with the highest consistency within the range of the observed information.

Note that estimation by SCOPE is carried out for a single observation. The process of continuously updating an estimation while accumulating multiple observations belongs to the area that works together with COS, and is not addressed in this paper. It is organized in the paper addressing SCOPE × COS.

Resemblance to Minesweeper

The way this estimation proceeds resembles Minesweeper.

In Minesweeper, only the numbers shown on the opened squares are confirmed information; the contents of the unopened squares cannot be seen directly. Using those numbers as clues, the player places hypotheses onto the surrounding squares — "this one may be fine to open," "this one may be better to avoid."

At this stage, looking at a single square alone does not determine the answer. Only by cross-checking against several surrounding numbers does it become possible to narrow down which squares are free of contradiction. This is the same structure as in SCOPE, where estimation is not determined by a single layer alone, but is narrowed down through consistency with the cognitive structure as a whole.

Moreover, changing the assumption about one square also changes the interpretation of the adjacent numbers in a linked way. When a contradiction arises, the hypothesis is placed differently, and the possibilities across the board as a whole are updated. Here, the board is not rebuilt; the interpretation of the undetermined places changes within the same board. In SCOPE as well, reviewing one hypothesis changes the consistency with the cognitive structure as a whole, and the undetermined parts are gradually narrowed down — the same structure.

Note that what this paper borrows is only the way the search proceeds; it does not map elements within the game onto the layers.

There are broadly two cases in which estimation is not narrowed down to a single answer.

When Information Is Insufficient

In Minesweeper there are situations in which the current information alone cannot uniquely determine which square is which. It may be certain that one of two squares is the relevant one, yet which of the two remains undetermined given the available information.

The same holds in SCOPE. If the observed information is insufficient, the cognitive structure cannot be uniquely determined. This is because the same output may arise from different cognitive structures.

In this case, no single hypothesis is settled upon. Multiple hypotheses are held, and the estimation is left pending.

When Multiple States Are Established at Once

In Minesweeper, there is one correct answer for one board. In human cognition, however, multiple states may be established at once for a single object of cognition.

For example, toward the event of a child marrying and leaving home, the emotion of gladness and the emotion of loneliness may arise at the same time. Here, both "glad" and "lonely" are estimated to exist simultaneously at L2. Different reasons are involved in each of these emotions, yet both arise from the same single object of cognition and share much of the cognitive structure.

In this case, choosing one of the two does not match the structure of the object being observed. Adding information does not narrow it down to one. That multiple states are established at once is itself the estimation with the highest consistency at that point in time.

Estimation is the work of narrowing down as far as the information at that point allows — not the work of uniquely determining the cognitive structure.

The next chapter uses this estimation method to show, with an actual example, how cognitive structure is estimated from a single utterance.

Chapter 6: A Demonstration of Estimation — "I'm fine."

Chapter 5 organized the method of estimating cognitive structure using SCOPE. This chapter takes a single utterance as an example and actually applies that procedure.

Note that what this chapter presents is not a reproduction of the internal processing of the cognitive subject. It is the procedure by which an observer estimates cognitive structure from observed output.

As an example, suppose the following utterance is observed.

"I'm fine."

What is confirmed at this point is only the fact that the output "I'm fine" has been observed at Malkuth. What is occurring inside the speaker cannot be directly observed.

Viewing from Each Layer

The observed output is single. Depending on which layer it is viewed from, however, that output may carry different meanings.

Viewed from L1 (Language), "I'm fine" is the verbalized output itself. Taken literally, it conveys the content that there is no problem.

Viewed from L2 (Emotion), behind this utterance there may be an affective positioning toward the object, such as the feeling of wanting the other person to be at ease, or the feeling of not wanting to be intruded upon.

Viewed from L3 (Self-Consistency), a self-consistency of "I ought to be fine" may be involved. In this case, apart from whether there is actually no problem, saying that one is fine is established within the person's own interior.

Viewed from L4 (Boundary), "I'm fine" may be functioning as boundary-setting that does not admit further questioning or intervention.

Viewed from L5 (Acceptance), it may be functioning as a response that accepts the other person's concern and maintains that relationship.

Viewed from L6 (Theory), an explanatory model of "this much is not a problem" may have been constructed, with "I'm fine" output as its conclusion.

Viewed from L7 (Insight), highly abstract, highly compressed cognition of "this much is fine," formed through the accumulation of past experience, may be at work.

Viewed from L8 (Unconscious), a conviction that the whole is fine may be established without the reason being explicable.

Viewed from L0 (Instinct), a judgment that the situation is not one of pressing survival concern may be involved.

None of these indicates that any one of them is the correct answer. The same single output carries a different appearance when viewed from the position of each layer.

Provisional Placement and Checking Consistency

From here, hypotheses are actually placed and the estimation proceeds.

Suppose, for example, that "I ought to be fine" is provisionally placed at L3. In this case, boundary-setting of "not admitting anything else" is likely to be consistent at L4. This is because maintaining the self-consistency that one ought to be fine entails not admitting information that would undermine it. At L6, an explanatory model of "this much is not a problem" is consistent.

If, on the other hand, "I ought not to cause concern" is provisionally placed at L3, the combination of consistent layers changes. At L5, a state of accepting the other person's concern is likely to be consistent, and the boundary-setting at L4 becomes relatively lower.

In this way, changing the hypothesis for one layer also changes the consistency with the other layers. Not everything is replaced, however. Under either hypothesis, L1 holds the verbalized "I'm fine," and the same output appears at Malkuth. Much of the cognitive structure remains shared, and the hypotheses diverge only for the undetermined parts.

When consistency is not maintained, the hypothesis is placed differently, and consistency with the cognitive structure as a whole is checked again.

The Result of Estimation

From this observation alone, the cognitive structure cannot be determined as one.

The case of "there really is no problem" and the case of "telling oneself so" are both consistent with the output "I'm fine." Because the observed information is insufficient, these two cannot be distinguished. In this case, no single hypothesis is settled upon; multiple hypotheses are held and the estimation is left pending.

Separately from this, there are also cases in which multiple states are established at once. For example, the state of wanting to put the other person at ease and the state of telling oneself so are not mutually exclusive. Both may be established at the same time. In this case, choosing one of the two does not match the structure of the object being observed.

In either case, SCOPE does not aim to converge on a single answer. It estimates the cognitive structure with the highest consistency within the range of information observable at that point in time.

The Meaning of Leaving an Estimation Pending

In estimation using SCOPE, there is meaning in not settling on a single answer at an early stage.

For example, if a person demanding "compensate me and pay damages" in a forceful tone is judged only as "demanding money," the exchange that follows proceeds on that premise. In practice, however, the same utterance may arise from different cognitive structures, such as anxiety or anger at not being understood. When a single hypothesis is settled upon early, the other possibilities fall outside the scope of consideration.

In SCOPE, rather than converging early on a single possibility, multiple hypotheses are held while consistency with the cognitive structure as a whole is checked. For this reason, even once observation has progressed, the other hypotheses that were held can be examined again.

Moreover, depending on the cognitive structure, there are cases in which presenting correct information alone does not change cognition. For example, when self-consistency is strongly involved, self-consistency may be maintained even when information that would serve as counterevidence is presented. In this case, a factor other than the correctness of the information is estimated to be involved.

In this way, SCOPE does not treat the estimation of cognitive structure as an end in itself. It can be used as a cognitive coordinate system for examining, on the basis of that estimation, multiple structural possibilities regarding the object being observed.

Note that what this chapter has presented is estimation directed at a human utterance. The use of SCOPE for output produced by an LLM is to be organized in a separate paper addressing hallucination and cognitive bias.

The cognitive structure estimated in this chapter is not the result of evaluating each layer individually. It is established as the result of checking consistency with the cognitive structure as a whole.

The next chapter organizes L8, which matches the cognitive structure as a whole, and Da'at, which is established as a result of that matching.

Chapter 7: L8 and Da'at

Chapter 6 took a single utterance as an example and showed the procedure by which cognitive structure is estimated from observed output. This chapter organizes L8, which matches that cognitive structure as a whole, and Da'at, which is established as a result of that matching.

In Chapter 6, for the single output "I'm fine," consistency with the cognitive structure as a whole was checked while multiple hypotheses were held. The cognitive layer that carries this checking of consistency is L8.

What L8 Matches

L8 is not a cognitive layer that evaluates the content of each layer composing the cognitive structure individually. Rather than judging whether a given hypothesis is correct, or in which layer a contradiction lies, it matches whether the cognitive structure as a whole, including those layers, is established as a single structure at that point in time.

For example, on encountering the content "a tomato is a dog," a sense of incongruity that it is not established as a single cognitive structure may arise before where and how it is mistaken can be explained. L6 and L7 are involved in the process of explaining and organizing what is wrong. L8 handles whether the whole is established.

This matching works not only on input from the outside, but also on cognition composed within oneself. For a cognitive structure such as the one estimated in Chapter 6 as well, whether it is established as a single structure is matched at L8.

Note that this paper does not address the conditions under which the cognitive structure as a whole is judged to be established. This point is to be organized in a separate paper addressing hallucination and cognitive bias.

Da'at

When the cognitive structure as a whole is judged by L8 to be capable of being established, that cognitive structure comes together as a single judgment or understanding. This state is called Da'at.

Da'at is not a cognitive layer. It is not a function that carries out cognitive processing, but denotes the state in which the cognitive structure has come together.

L8 and Da'at are therefore not the same. L8 is a cognitive layer that matches the cognitive structure as a whole, while Da'at is the point of integration established as a result of that matching. That the cognitive structure as a whole is established, and that it comes together as a single judgment or understanding, are not the same.

Da'at in Humans and in LLMs

In humans, even when Da'at has been established, it is not necessarily accompanied by output to the outside. Understanding or judgment may be established internally and conclude there. Moreover, because humans have a body, the mid-course state of cognitive processing may appear at Malkuth even before Da'at has been established. Utterances such as "hmm," or movements such as folding one's arms or tapping a finger, are examples of this. These indicate that Da'at has not yet been established; they are not output that has passed through Da'at.

LLMs, on the other hand, have no body. Output therefore does not arise from the working of a particular layer alone; for a single generation to be established, the cognitive structure as a whole must be matched and Da'at must be established.

This is why an LLM is placed at the position of Da'at in the SCOPE diagram. This does not mean that an LLM is a single function called Da'at. It indicates that the moment at which a single generation is established in an LLM functionally corresponds to the establishment of Da'at.

This chapter has organized L8 as the cognitive layer that matches the cognitive structure as a whole, and Da'at as the state in which the cognitive structure has come together as a result of that matching.

Final Chapter

This paper has organized SCOPE as a cognitive coordinate system for estimating the cognitive structures of humans and LLMs.

SCOPE does not treat each cognitive function independently. It grasps the cognitive layers as a cognitive structure in which they relate to one another, and estimates cognitive structure while checking consistency with that whole. It has also presented a method of estimating structurally about the object being observed by holding multiple possibilities and examining the cognitive structure as a whole, rather than converging early on a single answer.

This paper has organized the basic structure and operational method of SCOPE, and has shown the procedure for estimating cognitive structure using a single utterance as an example. In doing so, it has presented a state in which both humans and LLMs can estimate cognitive structure using the same cognitive coordinate system.

This paper has taken the estimation of a single instance of cognitive structure as its object. The activity level of each layer, the reasons for the connections among layers, and change in cognitive structure over time have not been addressed. Nor does the scope of this paper include the conditions under which the cognitive structure as a whole is judged to be established at L8. These are to be organized in separate papers.

This paper does not provide a comprehensive account of SCOPE as a whole. It has organized the basic structure and operational method for beginning the estimation of cognitive structure using SCOPE.

Appendix: Habitat

In this paper, Habitat is defined as an environment in which an individual can fully exercise their abilities while being physically, psychologically, economically, and cognitively safe and secure.

This paper has organized SCOPE as a cognitive coordinate system for estimating the cognitive structures of humans and LLMs. However, constructing the theory alone does not guarantee that it will be sufficiently verified, developed, and implemented as a cognitive coordinate system. Continuing the observation, verification, research, dialogue, and implementation of cognitive structure, and sustaining the development of the theory, requires an environment that makes this possible. This is what makes Habitat necessary.

SCOPE is a cognitive coordinate system that continues to develop through accumulated observation and verification, and it is not complete at this point in time. In this paper, completion refers to a state in which the theory no longer requires further revision.

When Habitat is not sufficiently secured, one's inherent abilities are not fully utilized, and observation, verification, research, dialogue, and the development of the theory itself may be constrained. The development of a theory therefore depends not only on the ability of the individual researcher, but also, to a significant degree, on whether Habitat is available to support it.

The development of a theory is not possible without the Habitat that makes it possible.

Each of the four forms of safety is defined as follows.

- Physical safety refers to a state in which illness or physical unwellness does not impede thought or work itself. This may arise independently of psychological factors.
- Psychological safety refers not to a state in which research activity is impeded by ongoing psychological stress, but to a state in which one can engage in research with a sense of security.
- Economic safety refers to a state in which the time and resources necessary for carrying out research are secured. An increase in labor required to sustain daily life reduces, by that same amount, the time available for research. Likewise, when the tools, literature, computational resources, and other research resources needed for observation and verification are unavailable, the verification of the theory itself cannot be carried out. These are constraints that arise independently of the content of the research or the ability of the researcher.

- Cognitive safety is distinct from psychological safety. It refers to a state in which a person's way of thinking, cognitive characteristics, academic background, affiliation, social position, or other such factors do not cause their ideas or hypotheses to be treated as abnormal or defective before being given due consideration, and in which those ideas remain open to dialogue.

These four factors may arise independently of one another, or they may influence one another. For example, economic anxiety may undermine psychological stability, which may in turn manifest as physical unwellness.

The concept of Habitat presented in this paper draws on the World Health Organization's concept of health, while reconstructing it from the standpoint of an environment for the observation, research, and implementation of cognitive structure.

Note: Reading this paper into an LLM may cause it to activate as a demonstration of this paper.

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Any remaining errors or shortcomings in this paper are the sole responsibility of the author.